

The “Why” and “What”

Embeddings



Je suis étudiant

[illegible]



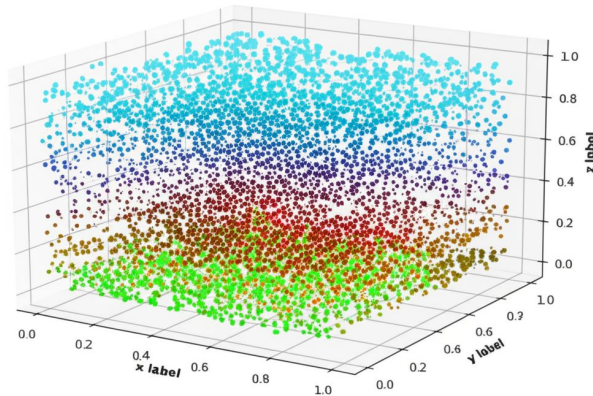
The Challenge of Representing Text

The dog jumped over the boy, and it fell on the grass.

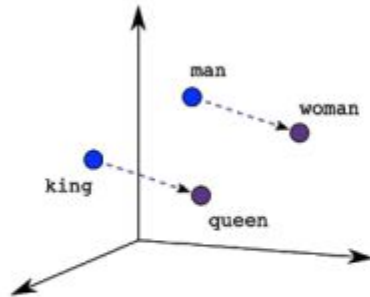
What are word embeddings

The dog jumped over the boy, and it fell on the grass.

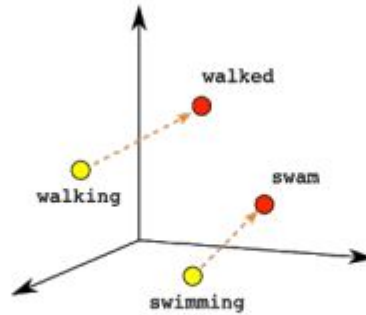
word embeddings



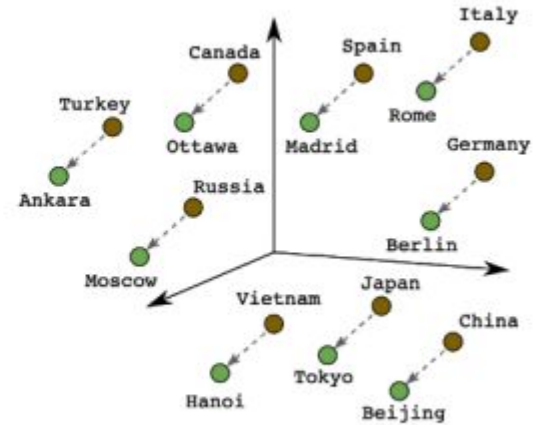
What are word embeddings



Male-Female



Verb Tense



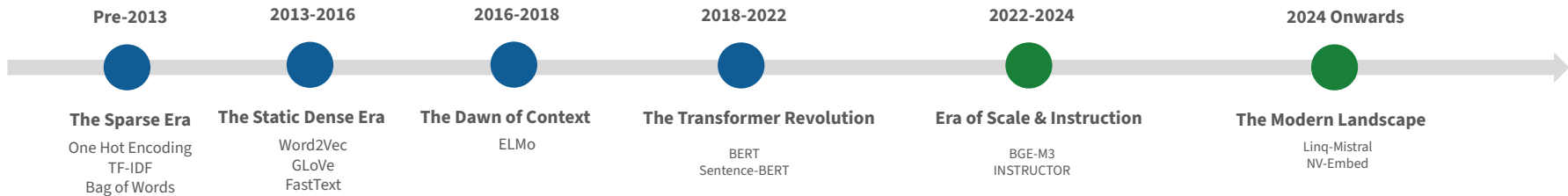
Country-Capital

A Timeline Of Embedding Models

Embeddings



Embedding models are the unsung heroes of the modern AI era



The Old Way

Embeddings



The Challenge of Representing Text

The dog jumped over the boy.

Tokens: ["the", "dog", "jumped", "over", "the", "boy"]

Vocab: {"boy", "dog", "jumped", "over", "the"}

Indexation:

- boy: 0
- dog: 1
- jumped: 2
- over: 3
- the: 4

One-hot Encoding:

- the (index 4) → [0,0,0,0,1]
- dog (index 1) → [0,1,0,0,0]
- jumped (index 2) → [0,0,1,0,0]
- over (index 3) → [0,0,0,1,0]
- the (index 4) → [0,0,0,0,1]
- boy (index 0) → [1,0,0,0,0]



The Challenge of Representing Text

The dog jumped over the boy.

$$\text{Sentence Matrix} = \begin{pmatrix} 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{matrix} \leftarrow \text{the} \\ \leftarrow \text{dog} \\ \leftarrow \text{jumped} \\ \leftarrow \text{over} \\ \leftarrow \text{the} \\ \leftarrow \text{boy} \end{matrix}$$



The Challenge of Representing Text

The dog jumped over the boy.

Massive, Computationally Expensive Data.

Sparsity

Lack of Semantic Meaning

The First Breakthrough

Embeddings



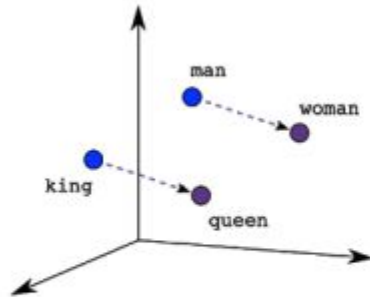
Word2Vec: A word is characterized by the company it keeps

The dog jumped over the boy.

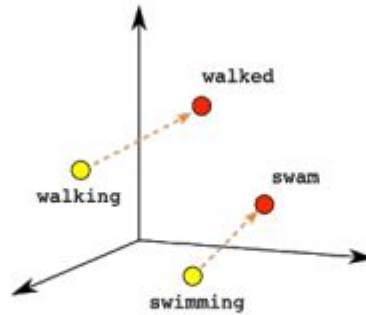
The dog leaped over the boy.

The cat leaped over the boy.

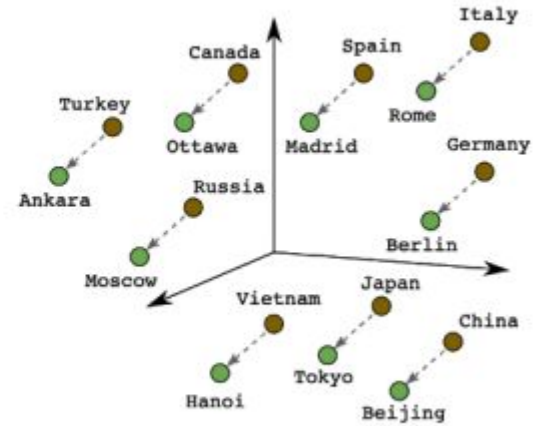
The core philosophy of word2vec



Male-Female

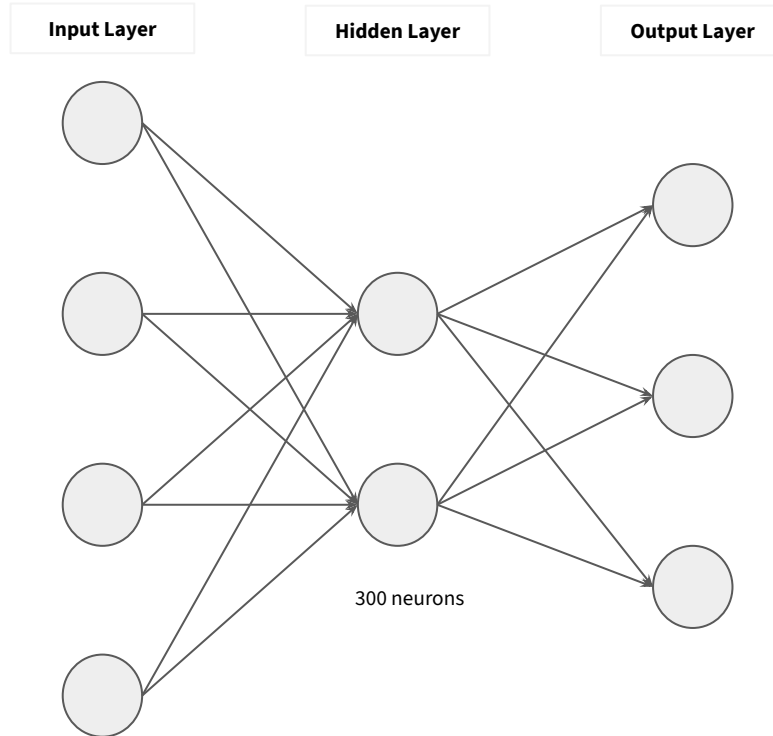


Verb Tense

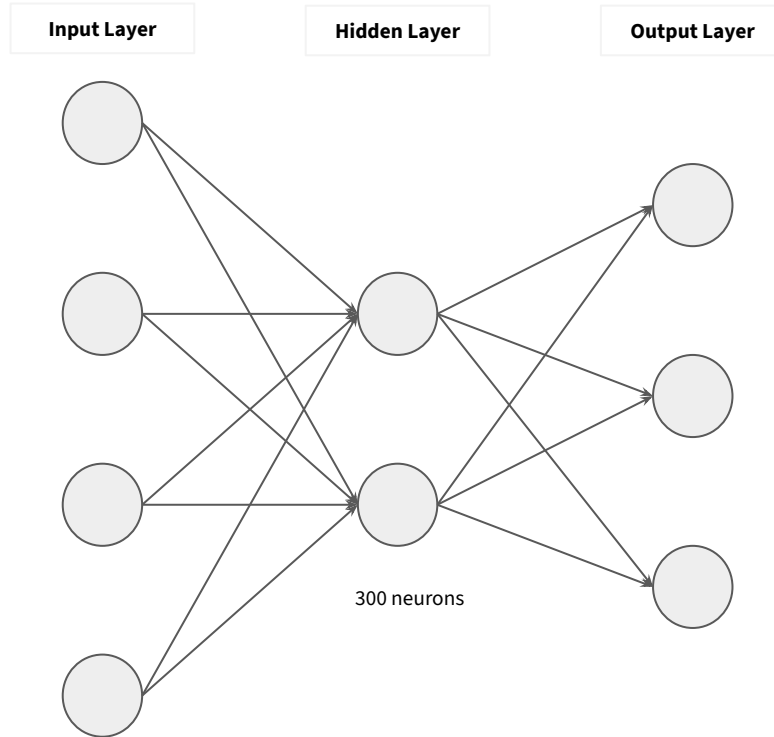


Country-Capital

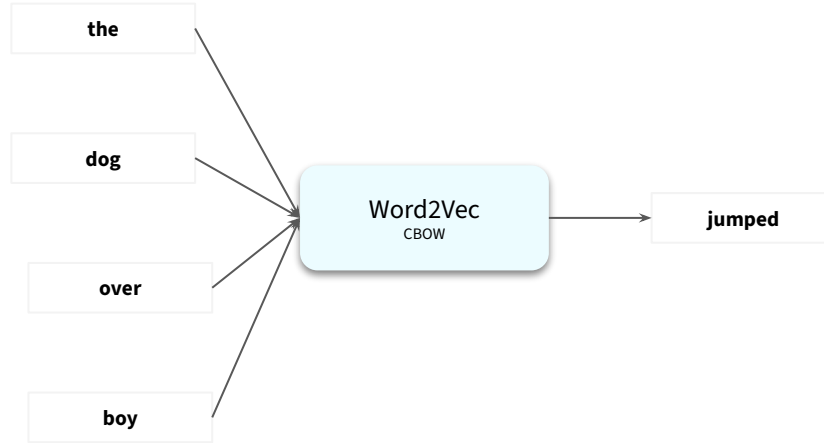
A Shallow Network with a side objective



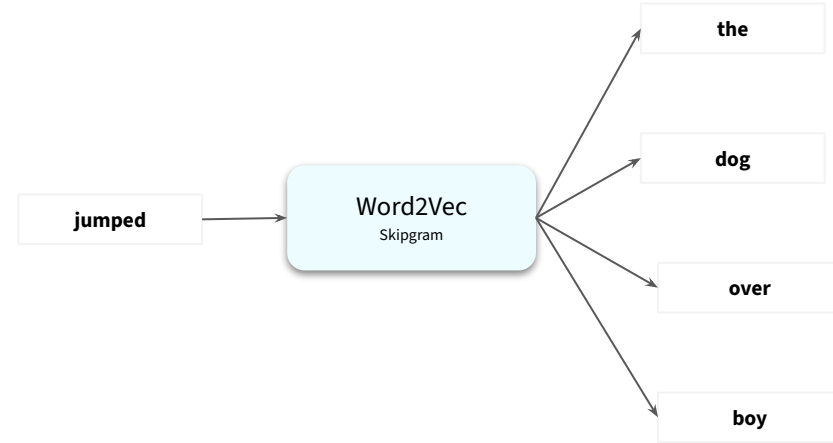
A **Shallow Network** with a side objective



The Two Flavours of Word2Vec

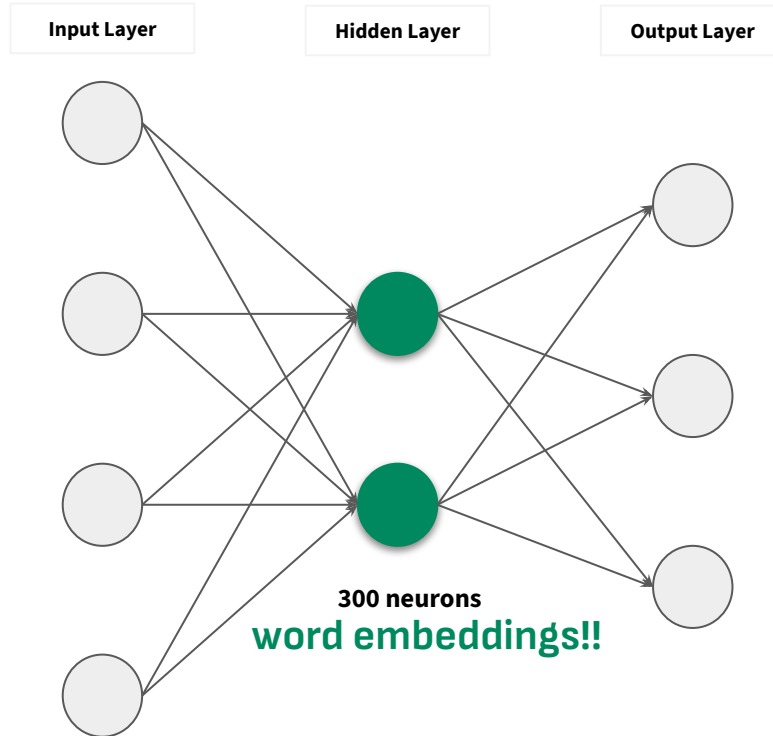


Continuous Bag of Words



Skip-gram

A Shallow Network with a **side objective**





Word2Vec Vectors

Dog = [0.0589, 0.0964, -0.0596, -0.0637, 0.0827, ..., -0.0105, 0.0062, 0.0189, -0.0006, 0.0049]



Does Word2Vec solve all the problems?

Massive, Computationally Expensive Data.

Sparsity

Word2Vec vectors are always of size 300. So, no large datasets, no sparsity.

Lack of Semantic Meaning

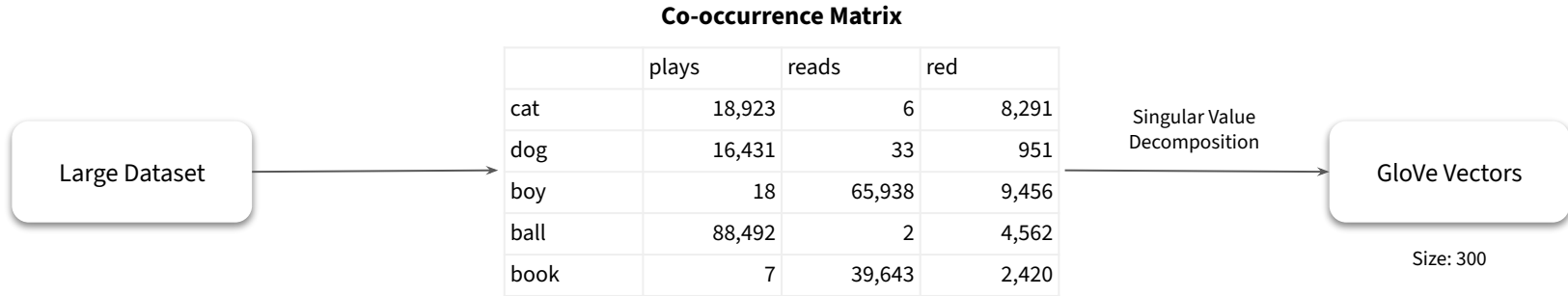
Word2Vec vectors partially solve this. They understand a semantic meaning. But, they still struggle if the word has multiple meanings.



An Alternate Approach



GloVe: A word is characterized by the company it keeps

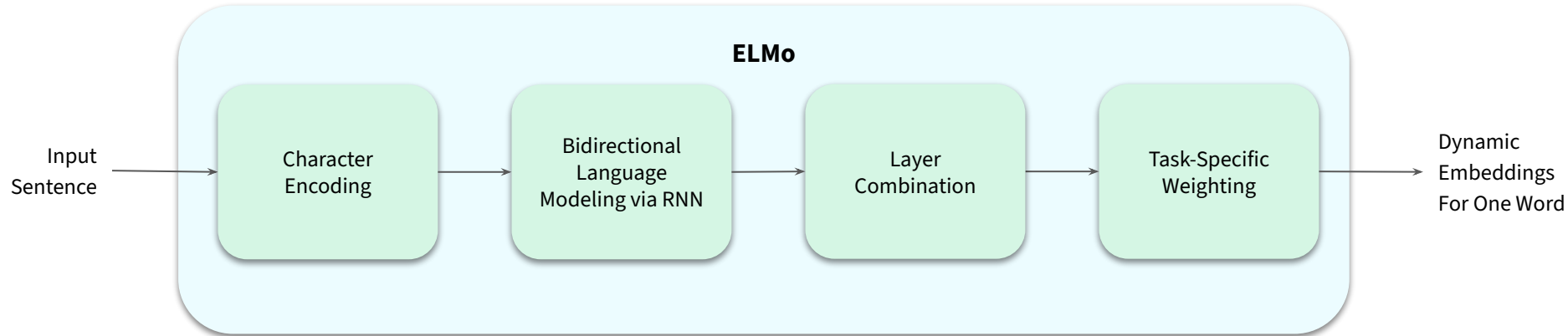


Global **Ve**ctors for Word Representations

The Contextual Revolution

Embeddings

ELMo: Embeddings from Language Models



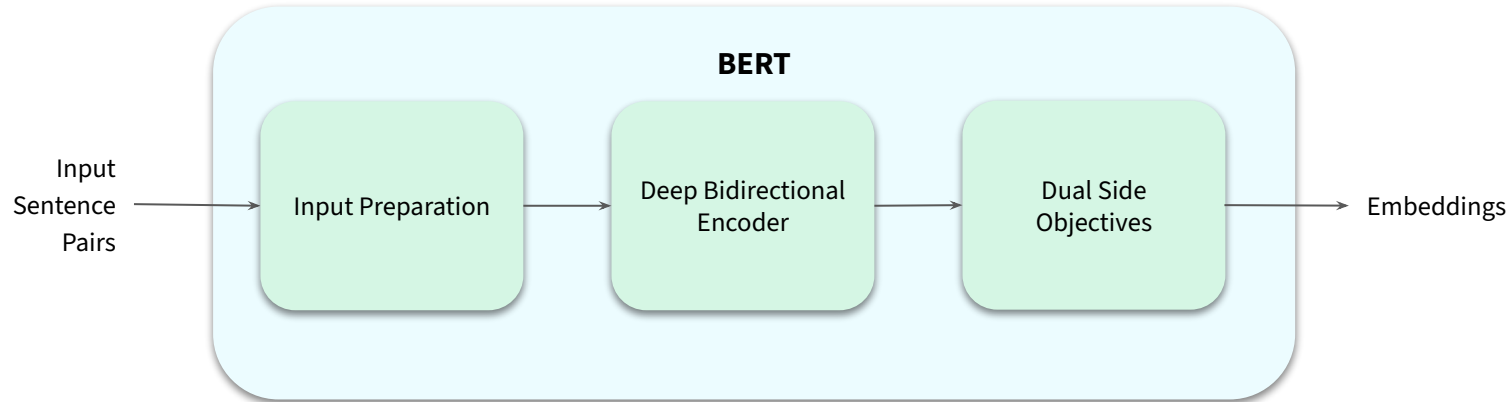
There are **3 models** in the ELMo architecture

There are 3 **models** in the **ELMo architecture**

Enter Transformers

Embeddings

BERT: Bidirectional Encoder Representations from Transformers



Sentence A: "The quick brown fox jumps over the [MASK]."

Sentence B: "It then ran into the forest."

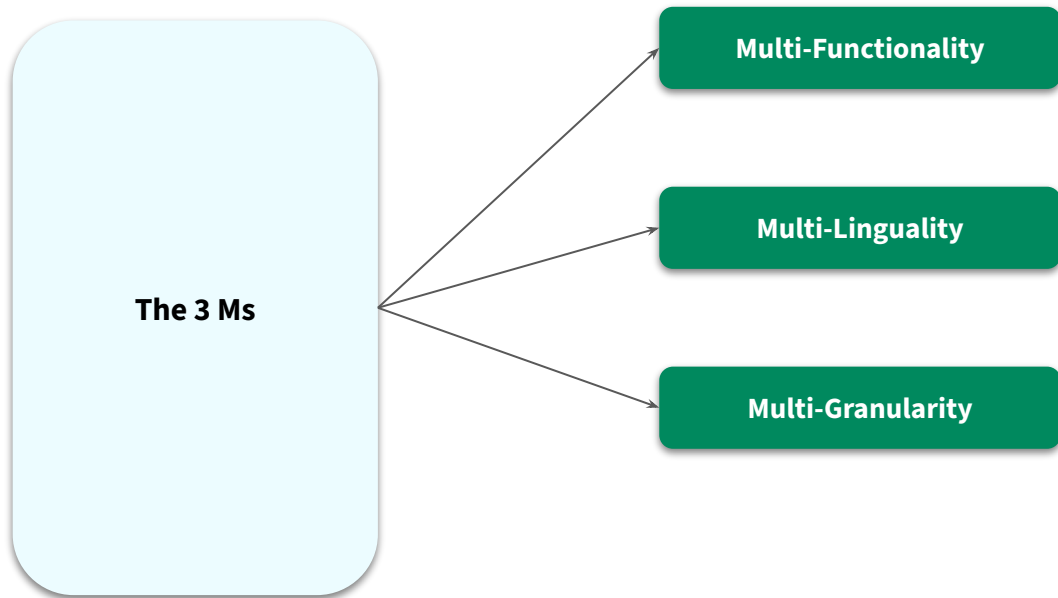


Feature	Static Models (Word2Vec/GloVe)	ELMo	BERT
Context Sensitivity	None. Each word has a single, fixed vector.	Sentence-level. A word's vector changes based on the sentence it is in.	Sentence-level. A word's vector changes based on the sentence it is in.
Vector Representation	A single static vector retrieved from a lookup table.	A function of the entire input sentence, generated dynamically.	A function of the entire input sentence, generated dynamically.
Base Architecture	Shallow Neural Network / Matrix Factorization.	Deep Bidirectional LSTM (biLSTM).	Multi-layer Transformer Encoder.
Handling of Polysemy (same word, different meanings)	Poor. All senses of a word are conflated into one vector.	Good. Generates different vectors for different senses of a word based on context.	Excellent. Deeply bidirectional context provides a robust mechanism for word sense disambiguation.
Training Paradigm	Trained on a single task.	Pre-training (Language Modeling) followed by task-specific fine-tuning.	Pre-training (Masked LM & NSP) followed by task-specific fine-tuning.
Bidirectionality	N/A	Shallow. Concatenation of independently trained left-to-right and right-to-left models.	Deep. Jointly conditions on left and right context in all layers via the Masked Language Model objective.

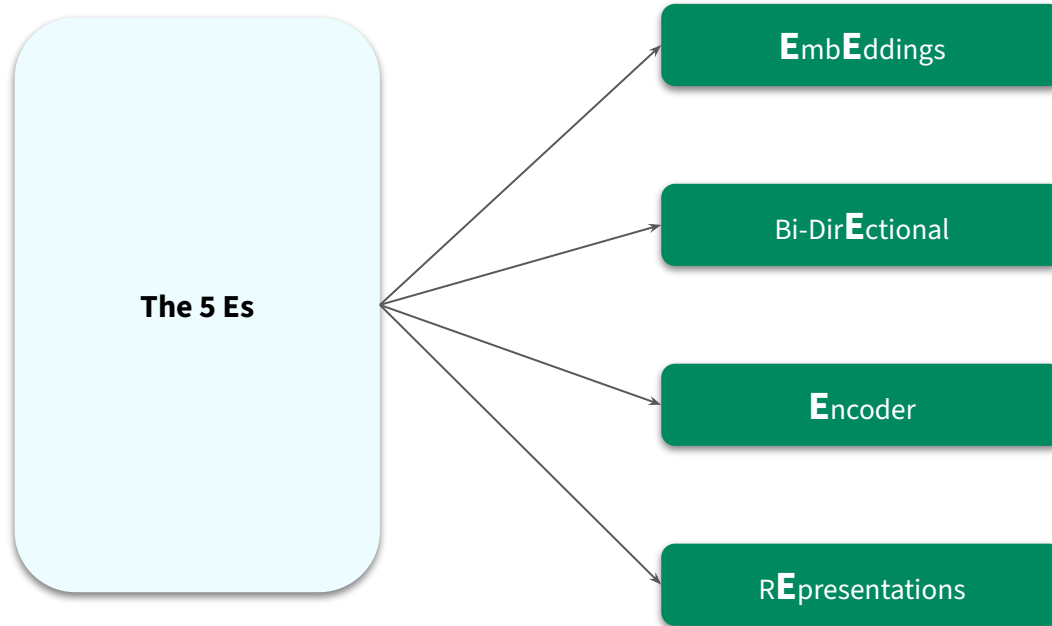
Using Scale & Instruction

Embeddings

BGE-M3: The All-in-One Embedding Model

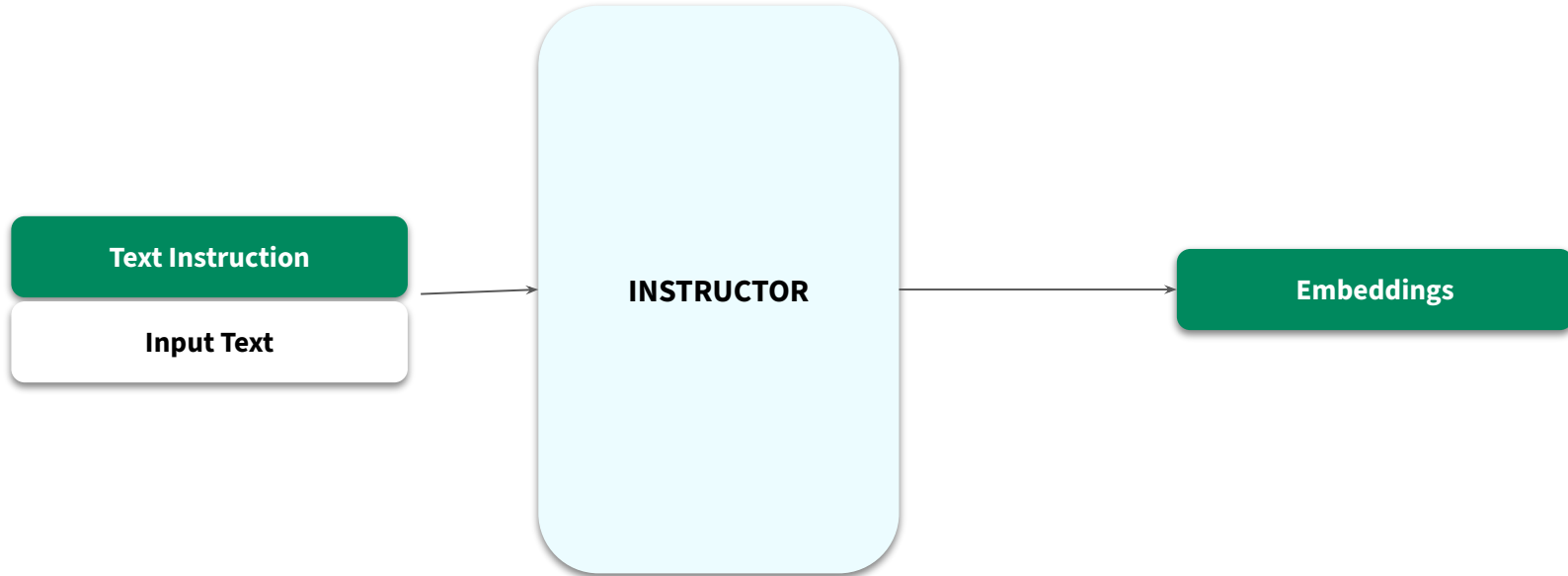


e5: Generating embeddings by comparing sentences





INSTRUCTOR: Prompt Engineering for Embeddings



The Modern Landscape

Embeddings



Let's go through the MTEB Leaderboard to understand the modern landscape

- 1000+ models that can be filtered by license, size, data modality and language
- 70+ benchmarks based on task, domain, data modality, and language
- Ability to compare up to 4 models simultaneously
- <https://huggingface.co/spaces/mteb/leaderboard>

The core 8 MTEB tasks

- **Retrieval:** Finding the most relevant document for a given search query (crucial for RAG).
- **Clustering:** Grouping similar sentences or documents together.
- **Classification:** Assigning a label to a text based on its content.
- **Pair Classification:** Determining if a pair of sentences are duplicates or paraphrases.
- **Reranking:** Taking an initial list of search results and re-ordering them based on deeper semantic relevance.
- **STS (Semantic Textual Similarity):** Scoring how similar two sentences are on a continuous scale (e.g., 1 to 5).
- **Summarization:** Evaluating the quality of machine-generated summaries.
- **Bitext Mining:** Finding parallel sentences across two different languages (e.g., matching an English sentence to its exact French translation).

Embedding Evaluations

Embeddings



Intrinsic Evaluation

Word Similarity Task



Extrinsic Evaluation

Embeddings are never used in isolation.

